Beyond the noise - feature selection is all you need!

“atleast locally”

danzell

agenda

- Intro
- my data science journey
- shadow null importance
- diagnostic scores
- possible improvements

Daniel /danzell

ML ENGINEER

NUMERAI

@danzell

kaggle

@danzel

my data science journey

kaggle

Porto Seguro‘s
Safe Driver Prediction

mjahrer

Denoising
Auto
Encoders

#	△	Team	Members	Score	Entries	Last	Solution
1	—	danzel		0.69381	46	3y	
2	▲1	Ren		0.69485	25	3y	
3	▲2	Fatih		0.69496	58	3y	
4	▼2	Dave E		0.69500	40	3y	
5	▼1	杰少		0.69509	47	3y	
6	▲3	Andy (I-Fan Chen)		0.69517	101	3y	

x2
x4

NUMERAI

long story short

- DAE work best on “clean” data
- No engineered features
- No noisy features

how to null feature importance

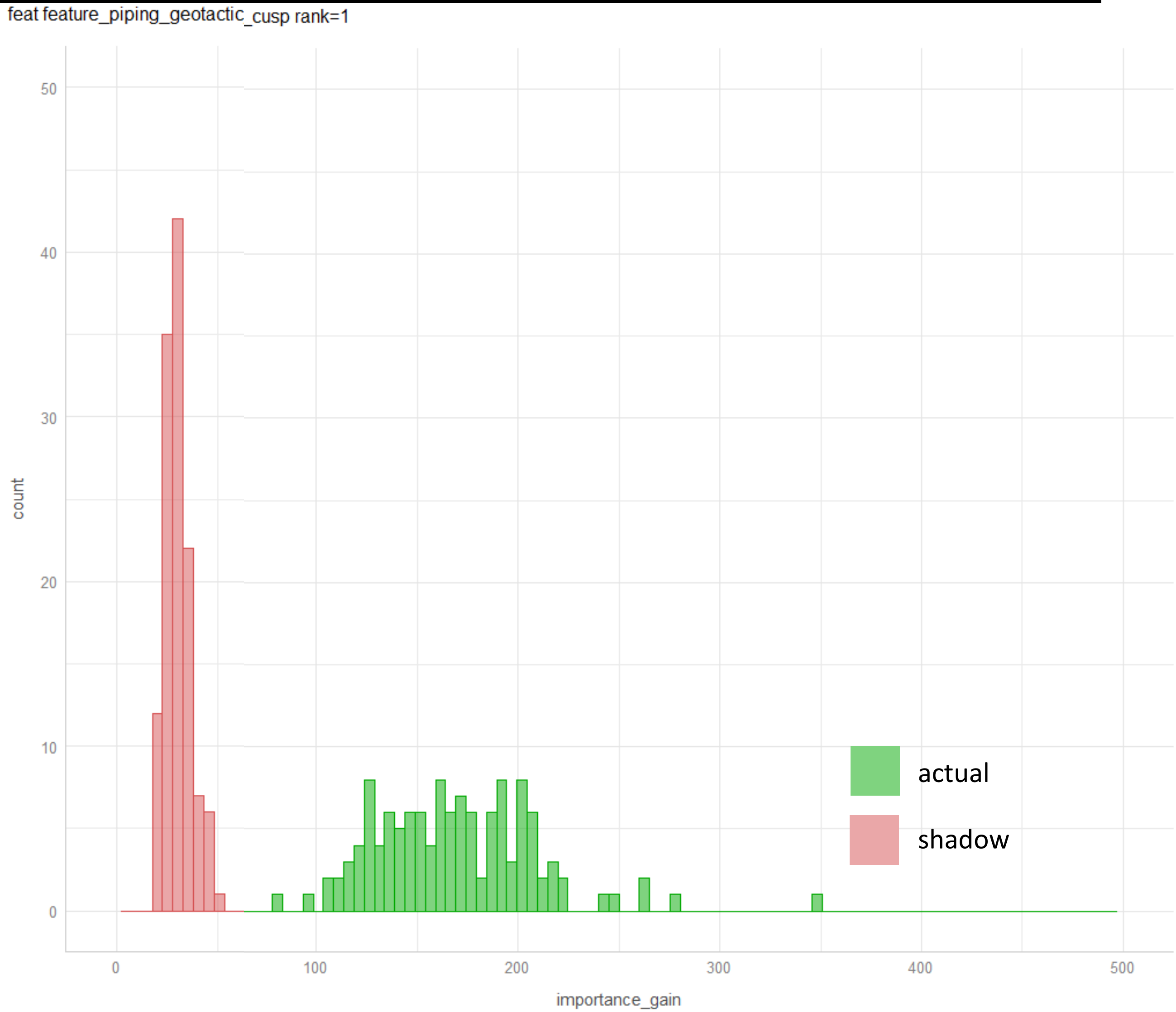
- clean target
- shuffled target

- train a tree-booster model
- calculate importance gain
- save results

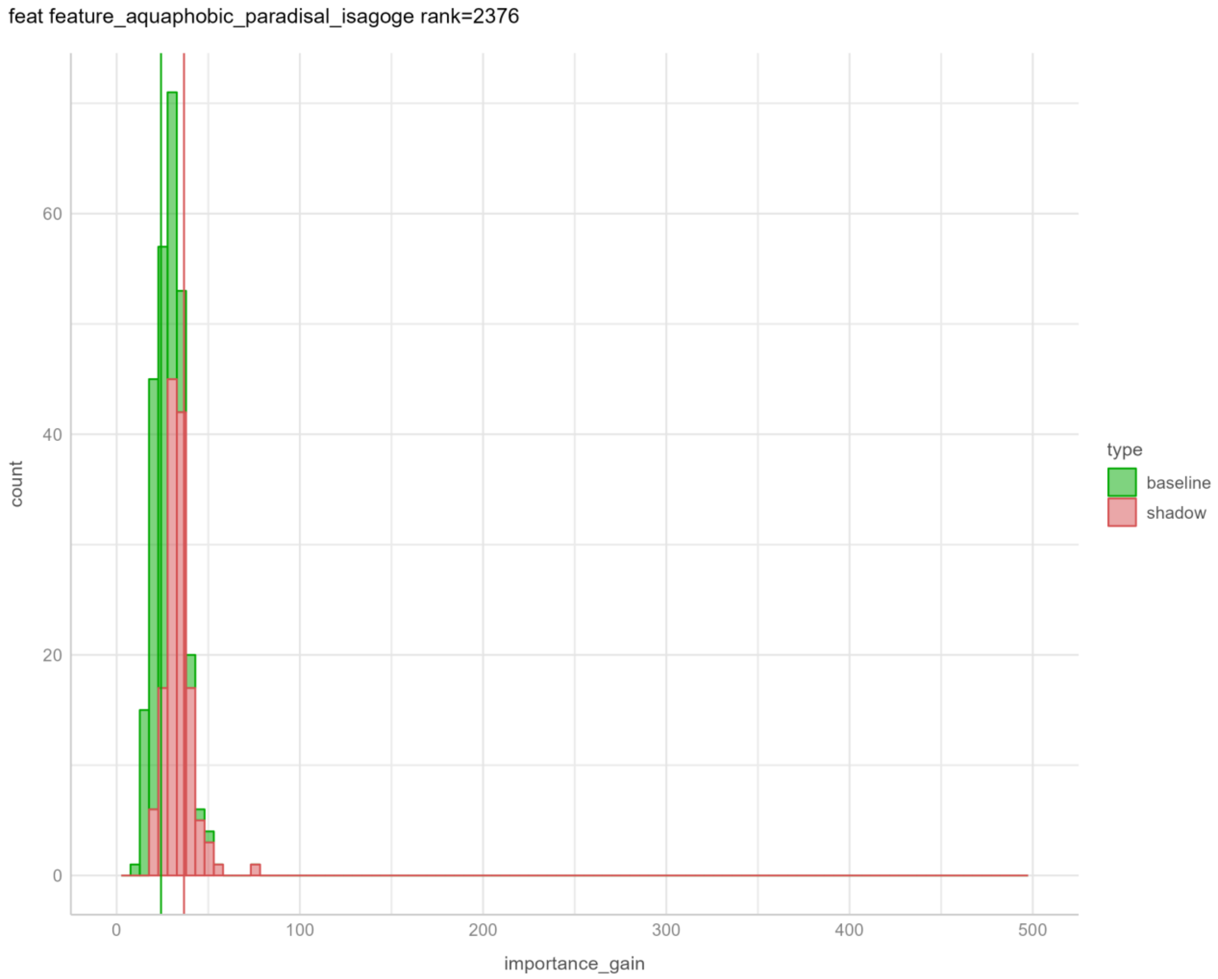
```
N_ESTIMATORS = 2000
LR = 0.01
MAX_DEPTH = 5
COL_SAMPLE = 0.35
```

change seed
repeat

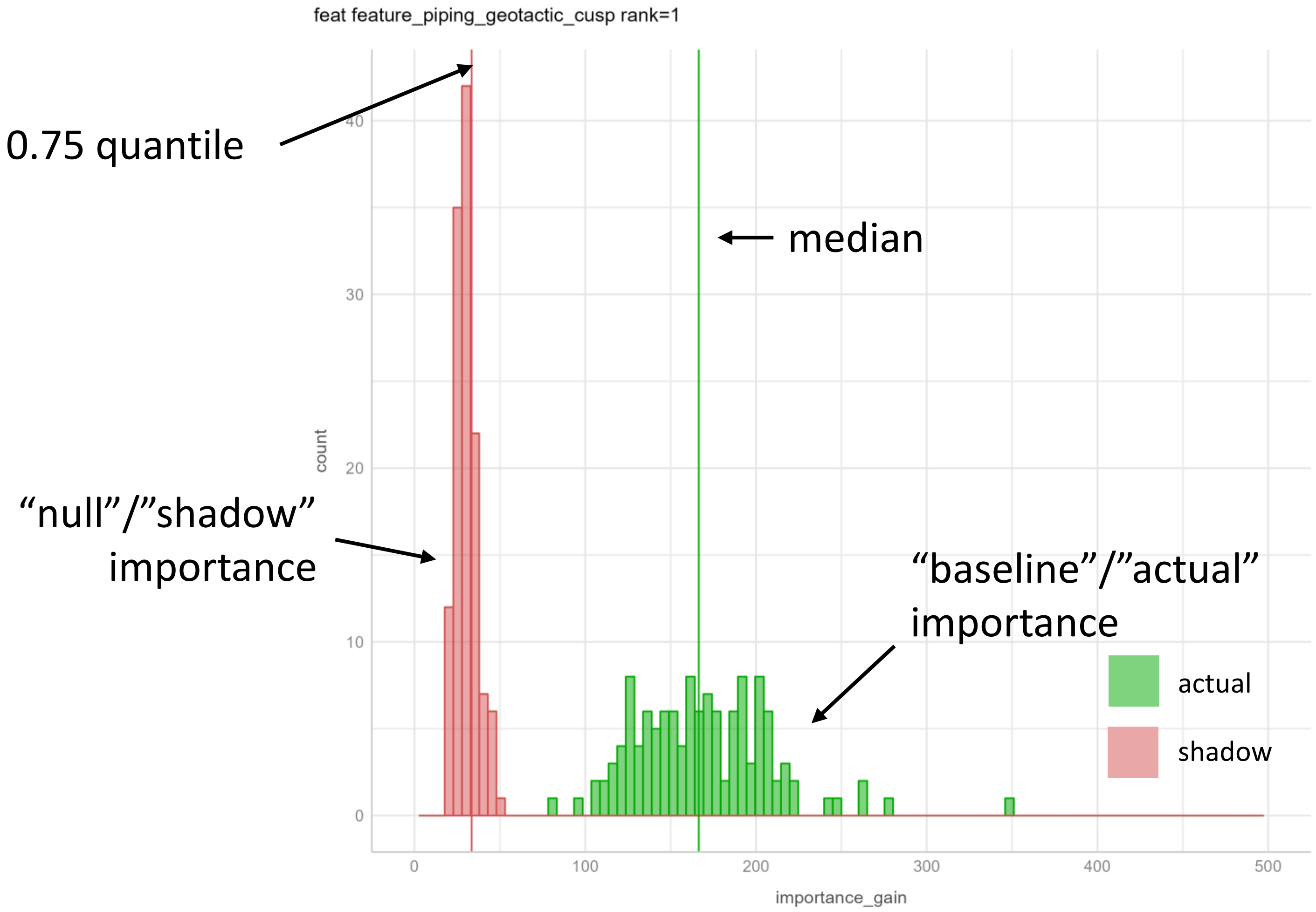
good vs bad features



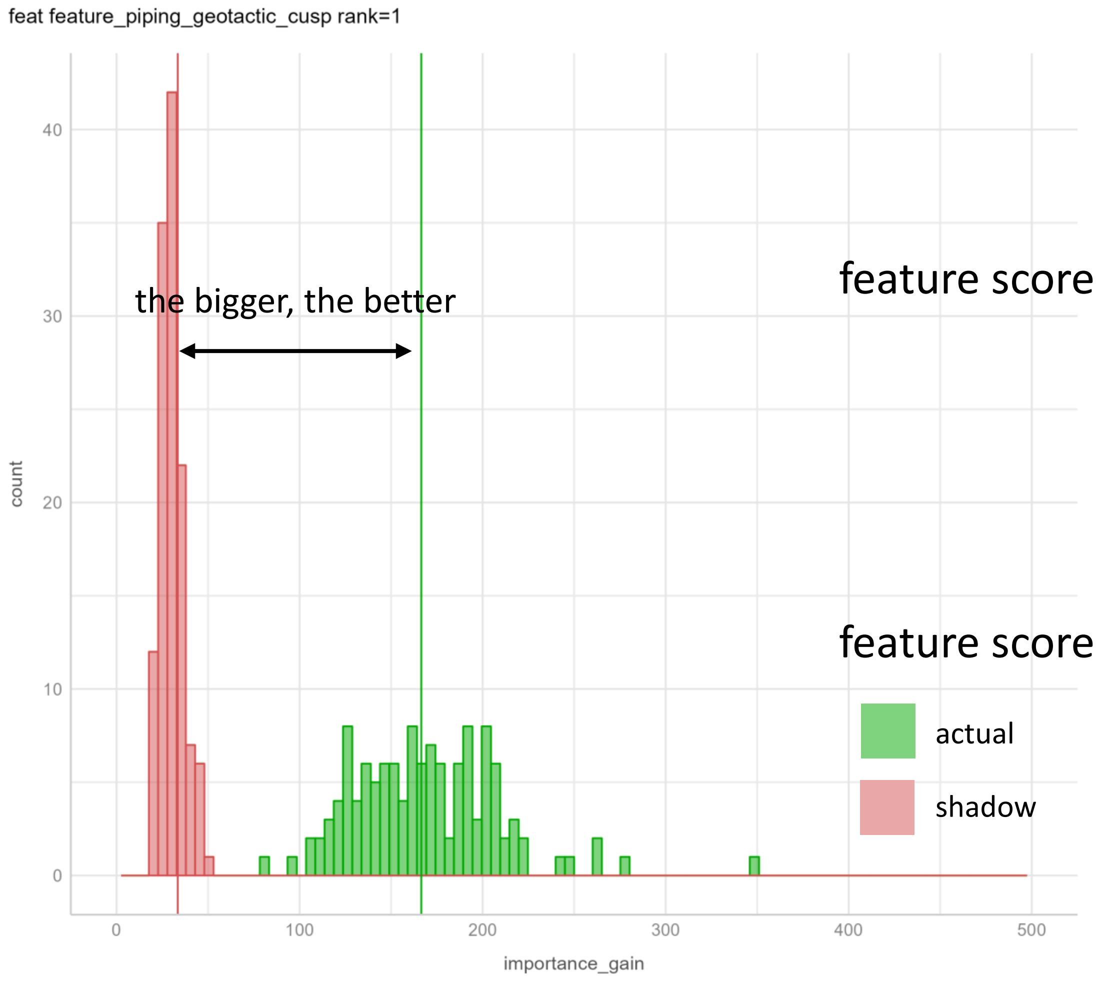
good vs bad features



good vs bad features



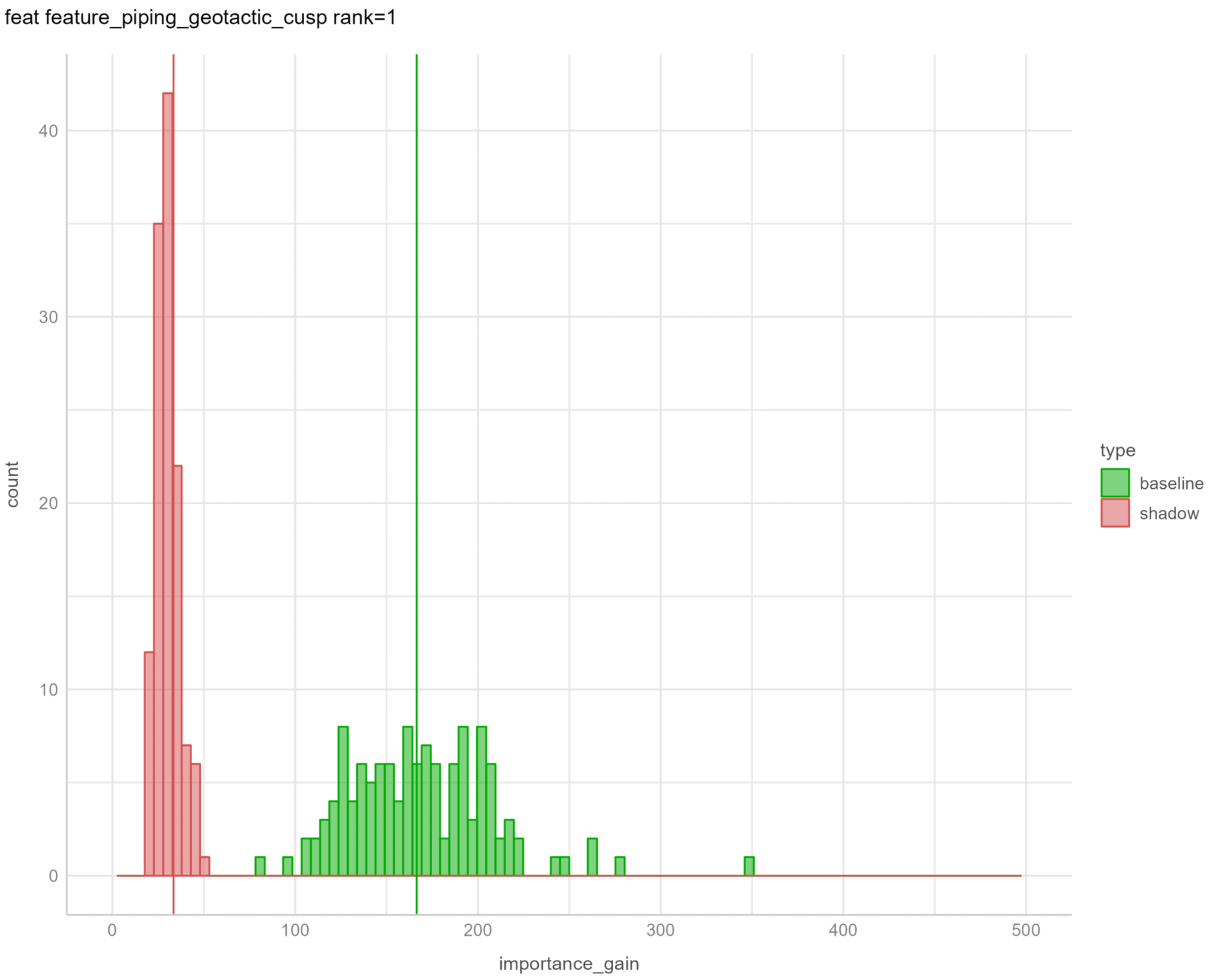
good vs bad features



$$\text{feature score} = \frac{\text{median actual}}{\text{0.75 quantile shadow}}$$

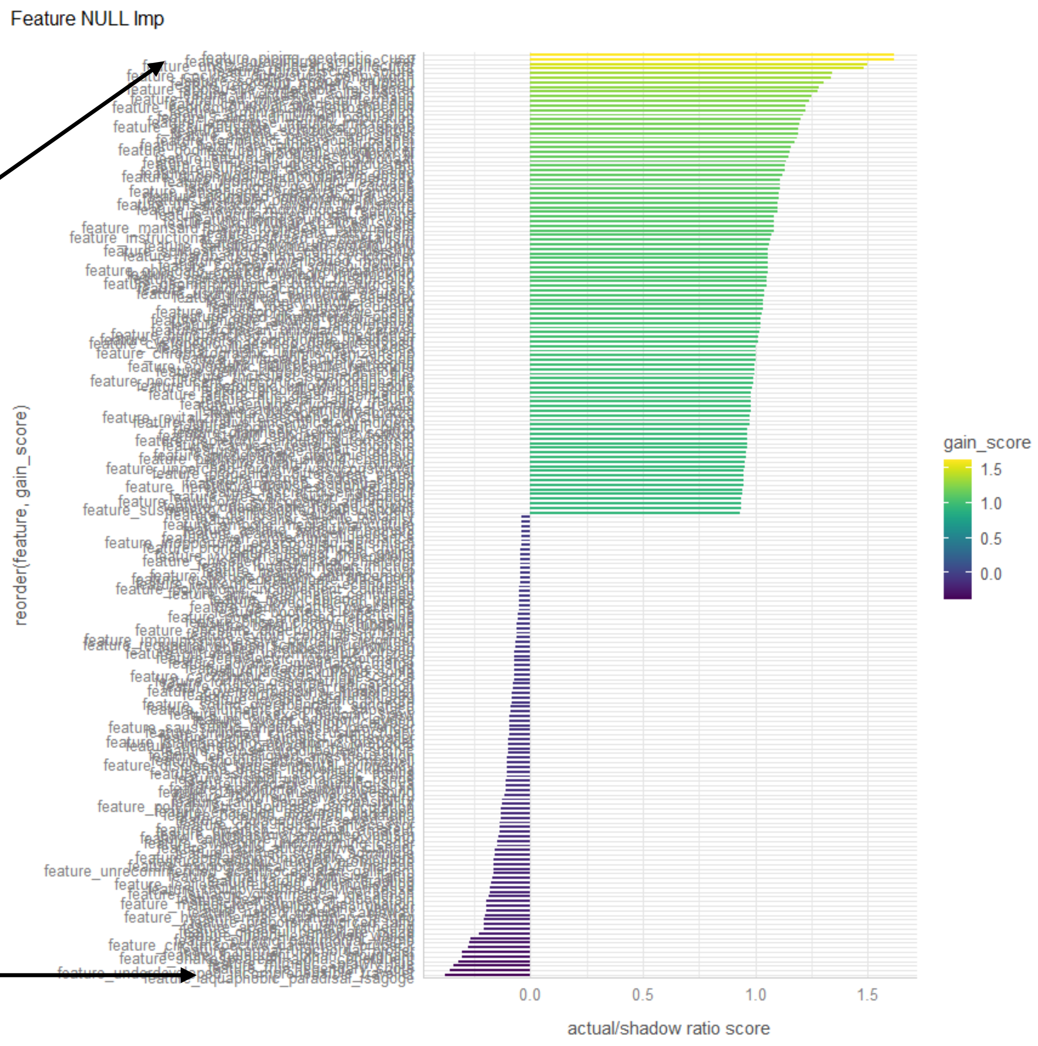
$$\text{feature score} = \log\left[\frac{\text{median actual}}{\text{0.75 quantile shadow}}\right]$$

all v43 features



Top/bottom 100

feature_piping_geotactic_cusp

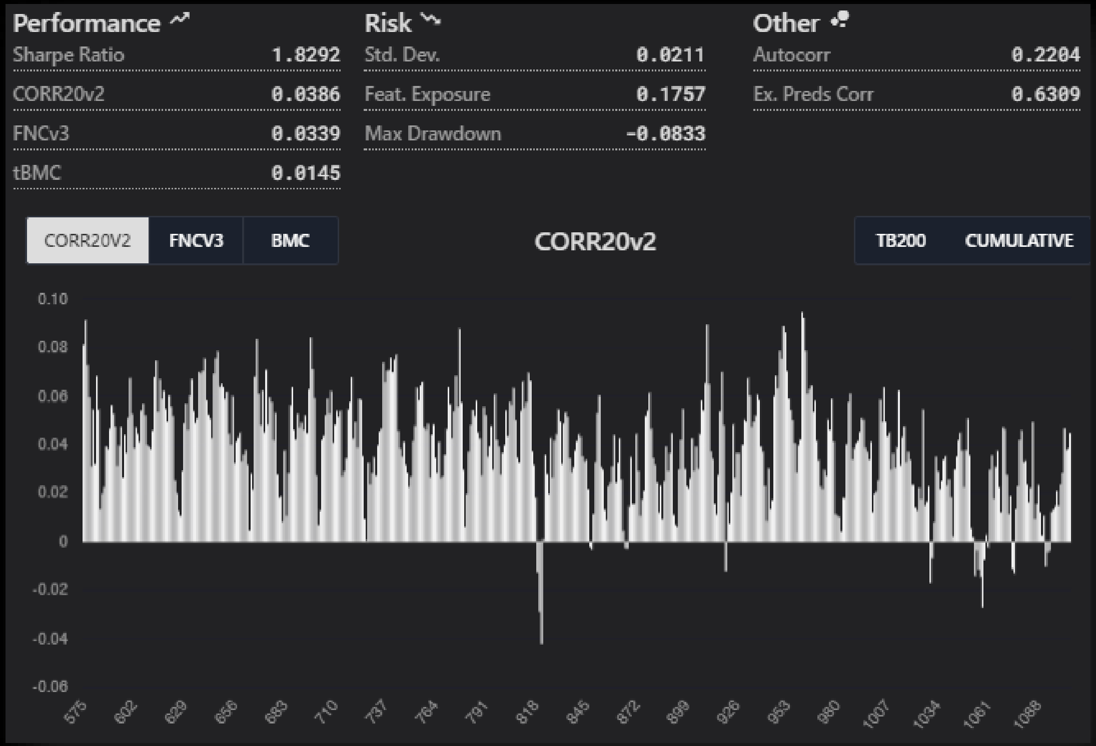


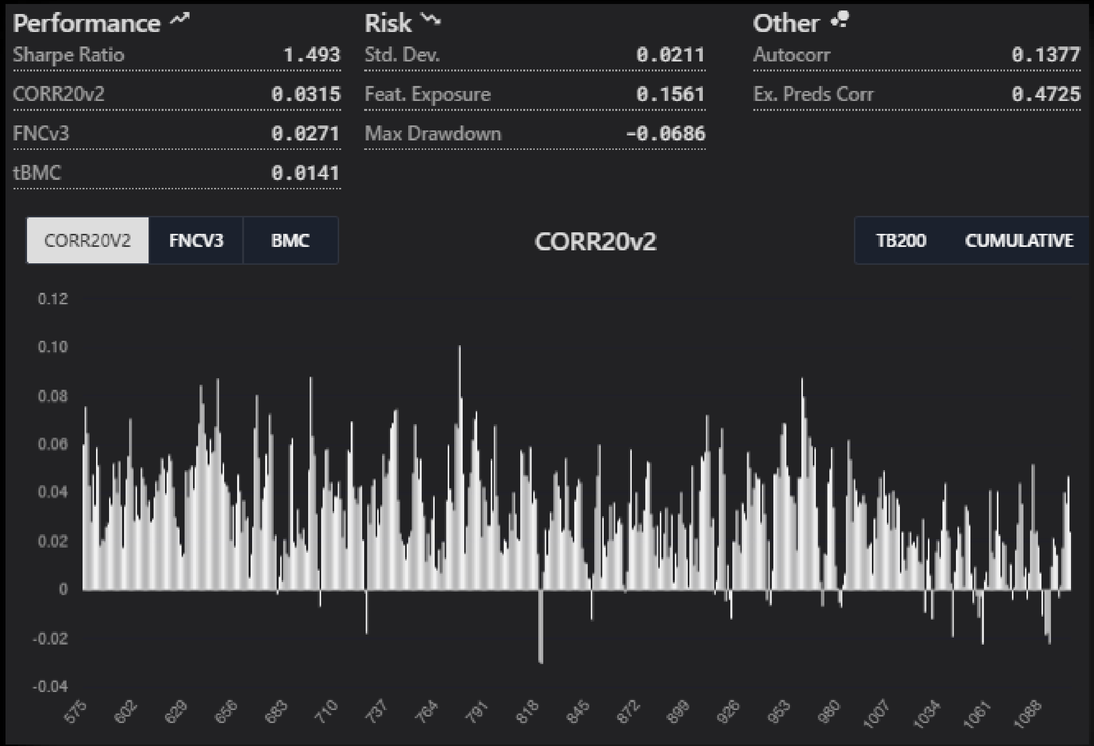
feature_true_legendary_shote

define cutoff

~ 200, ~500

example scores (trEra 1-560 valEra 575-11xx)

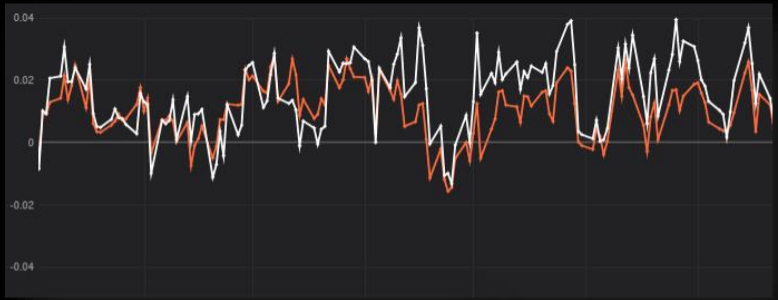


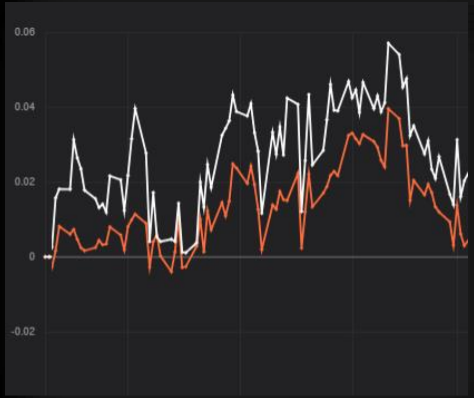


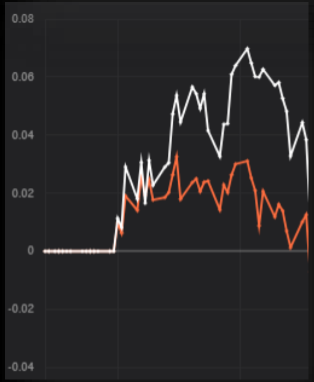
improvements

- combine calculated scores + corr/bmc/sharpe/ …
- use different algo/ hyperparameters
- change score calculation kullback leibler divergence
- check if there is a difference in training vs test
- change targets

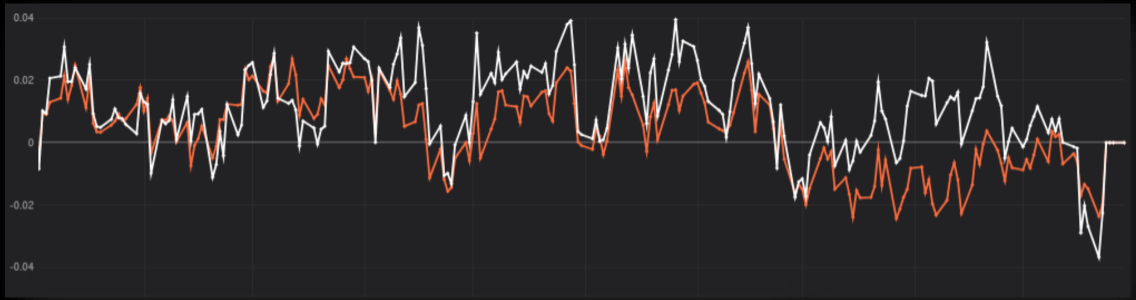
example models

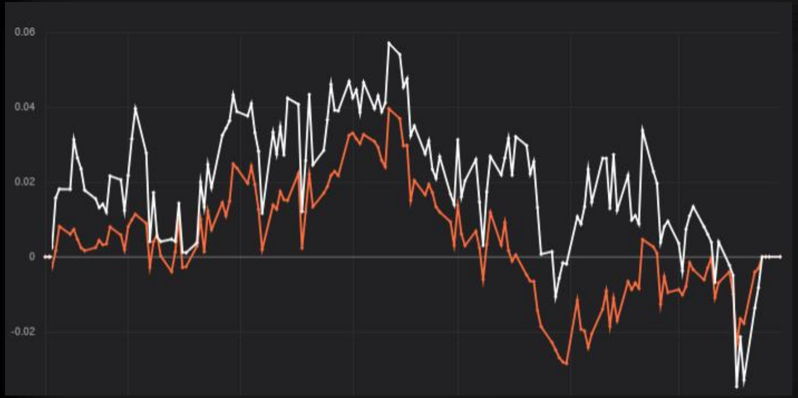


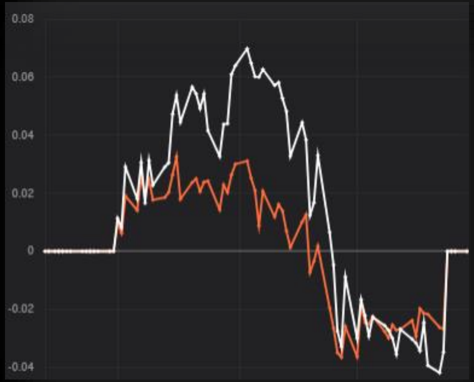




example models







Fin.